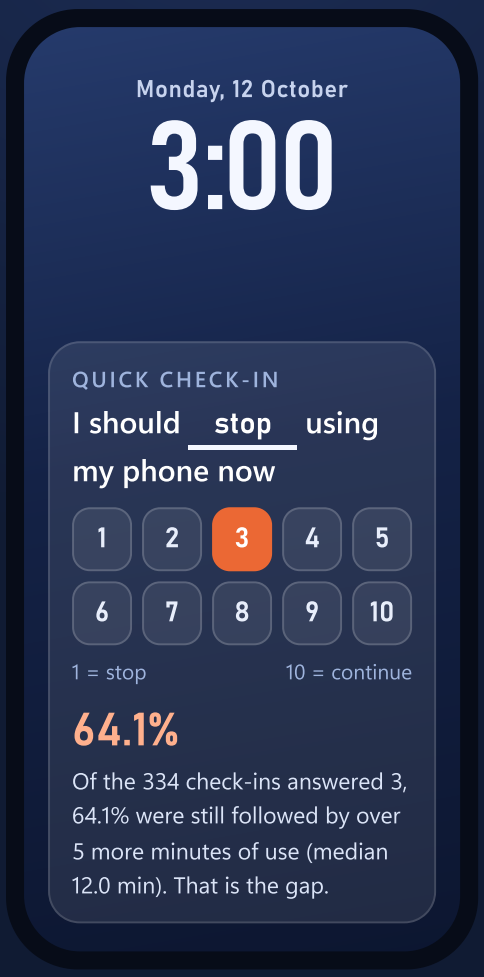


NUDGEWELL · GROUP 6 · INTERACTIVE SUPPLEMENT

The gap between "I should stop" and actually stopping.

That gap is what NudgeWell predicts. The phone shows the real check-in question that 37 adults answered 5,772 times in the AwareMind study (Zheng & Choe, 2026b). Choose an answer to see what people who gave it actually did next.

14.9%

of all check-ins were high-gap moments: leaning "stop", then over 5 more minutes of use

33.7%

of check-ins between midnight and 6 am, against 12.6% from 9 am to 6 pm

Supplement to **NudgeWell: Literature, Data & Technology Review** · Digital Addiction Capstone, Samsung Innovation Campus AI, Cohort 18, in partnership with Yeşilay · 12 October 2026

Group 6: Yusuf Ersel Kara · Bahadır Maral · Çağla Eryılmaz · Fatih Mehmet Menteşe · Damla Nur Öcal

NUDGEWELL IN ONE MINUTE

Predict the unwanted moment, then help once or stay quiet

Problem
Time limits interrupt purposeful use and miss unwanted use.

Target
The intention–behavior gap: the person says "stop", then keeps going.

For whom
Adults who use their phones heavily and want to cut down.

Data
AwareMind: 5,772 check-ins from 37 US adults, July–October 2025.

Phone use
Android usage events, read on the device

Context
Time, app type, minutes in session, earlier sessions

Gap model
Starts global, then learns the person

Decide
Confident, with prompt budget left? Otherwise stay quiet

One suggestion
From the user's own list; dismissed with one tap

WHAT THE DATA SHOWS

Five things the AwareMind data shows

The analysis uses all 5,772 in-session check-ins from 37 US adults (Zheng & Choe, 2026a). A **high-gap moment** means leaning toward "stop" (an answer below 5.5) and then using the phone for more than 5 more minutes. Figures are numbered as in the report.

14.9%

of check-ins were high-gap moments (859 of 5,772), matching the paper's published figure exactly

REPORT: INSIGHT 1 · FIGURE 1

Most check-ins lean toward continuing, but the gap is a substantial minority. 72% of answers leaned toward continuing, and 29.4% were a firm 10. Further use was heavily skewed: a median of 5.1 minutes, while 5.2% of check-ins ran past an hour.

SUPPLEMENTARY ANALYSIS

Leaning toward "stop" barely changed what happened next: 52.6% of those check-ins were followed by more than 5 more minutes of use, against 50.9% when leaning toward "continue". Only a firm 1 usually ended the session (26.7%). After an answer of 3, the median further use was 12 minutes.

What happened after each answer (all 5,772 check-ins)

Answer	1	2	3	4	5	6	7	8	9	10
Check-ins	311	193	334	430	364	527	491	637	790	1,695
Over 5 more minutes	26.7%	50.3%	64.1%	61.9%	54.7%	46.9%	45.0%	47.3%	50.6%	55.5%
Median further use (min)	1.3	5.1	12.0	9.6	6.2	4.6	3.9	4.4	5.1	5.9

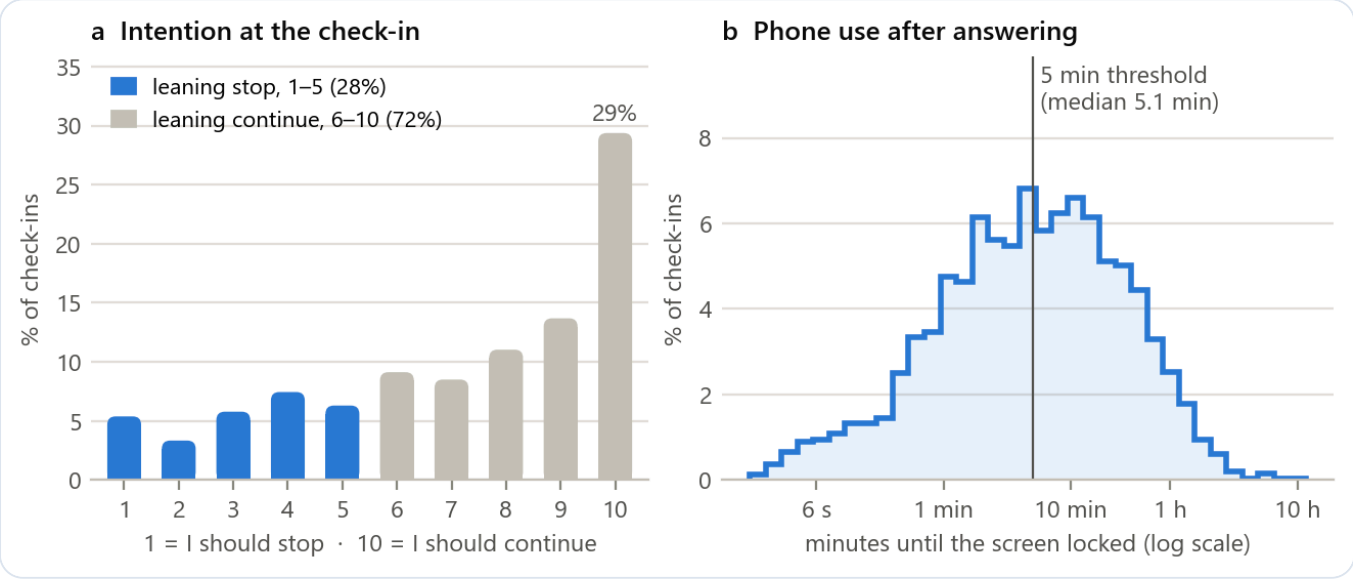


Figure 1. (a) Stated intention at the check-in. 28% of answers leaned toward stopping. (b) Phone use after answering; the vertical line marks the 5-minute threshold used to define high-gap moments.

33.7%

high-gap share between midnight and 6 am (n = 395), against 12.6% from 9 am to 6 pm

REPORT: INSIGHT 2 · FIGURE 2

The gap peaks at night, reaching 45% at 3 am (n = 49, with a wide confidence interval). Weekends (15.4%) barely differ from weekdays (14.7%). This matches the finding that usage profiles with lower well-being are marked by night-time sessions (Katevas et al., 2018). Time of day is known before the outcome, so a real-time model can use it.

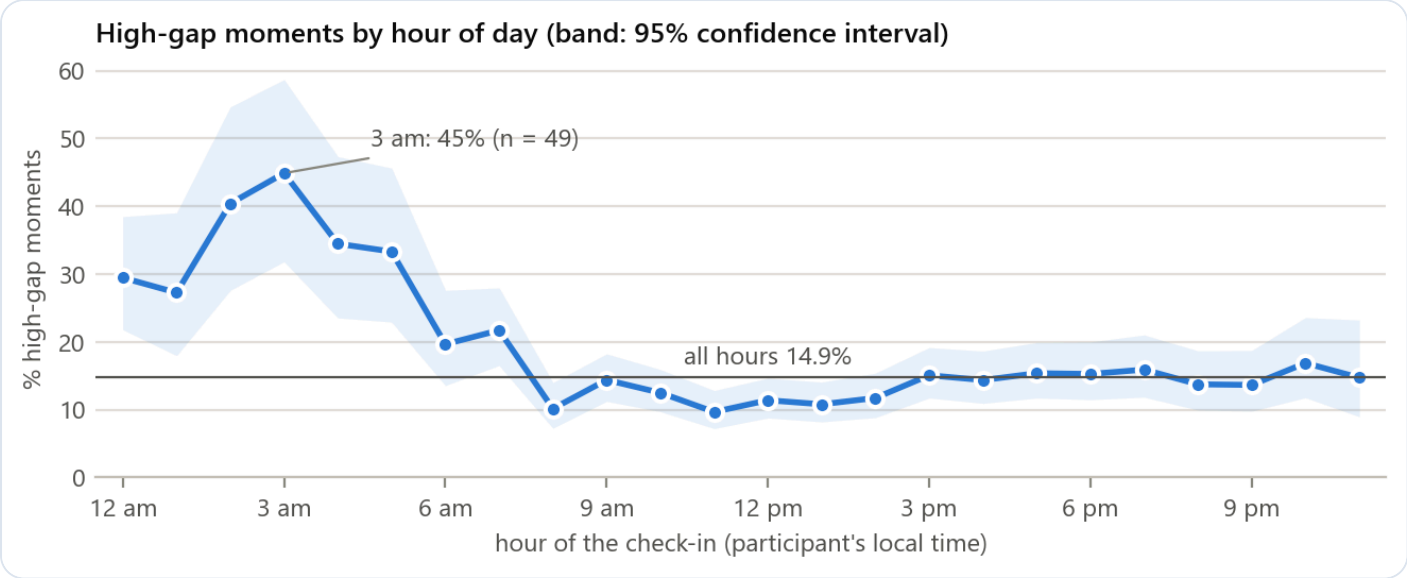


Figure 2. Share of high-gap moments by hour of the check-in (participant's local time), with 95% Wilson confidence intervals. Each hour has between 47 and 456 check-ins.

23.7%

high-gap share in social media apps (n = 908), against 10.6% in messaging

REPORT: INSIGHT 3 · FIGURE 3

Social media carries the largest gap, followed by media and video (18.4%) and browsers (16.2%). System tools are lowest (4.4%). This agrees with one sec users choosing to restrict social media most often (Haliburton et al., 2024). The app category is missing for 12.4% of check-ins.

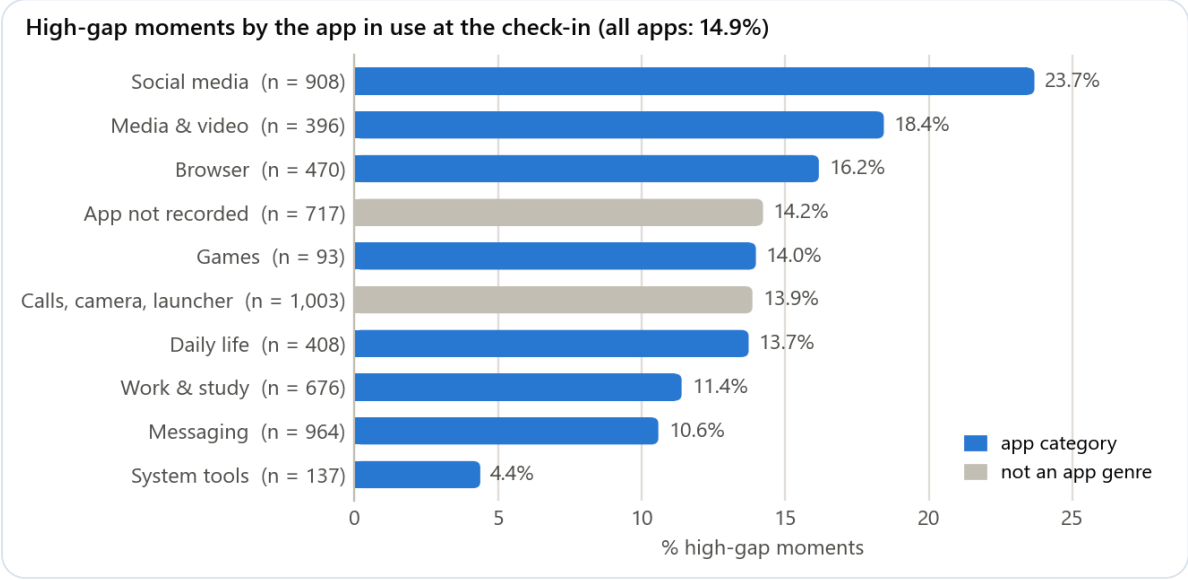


Figure 3. Share of high-gap moments by the category of the app in use at the check-in. Gray bars are technical groups rather than app genres.

13% → 22%

high-gap share from the first 30 seconds of a session to more than 10 minutes in

REPORT: INSIGHT 4 · FIGURE 4

The gap grows as a session goes on (13.4% → 21.6%), so elapsed time is a useful real-time signal. The participants are heavy users: at least 3.7 hours of screen time on the median participant-day, rising with age (3.0, 3.6 and 4.6 hours).

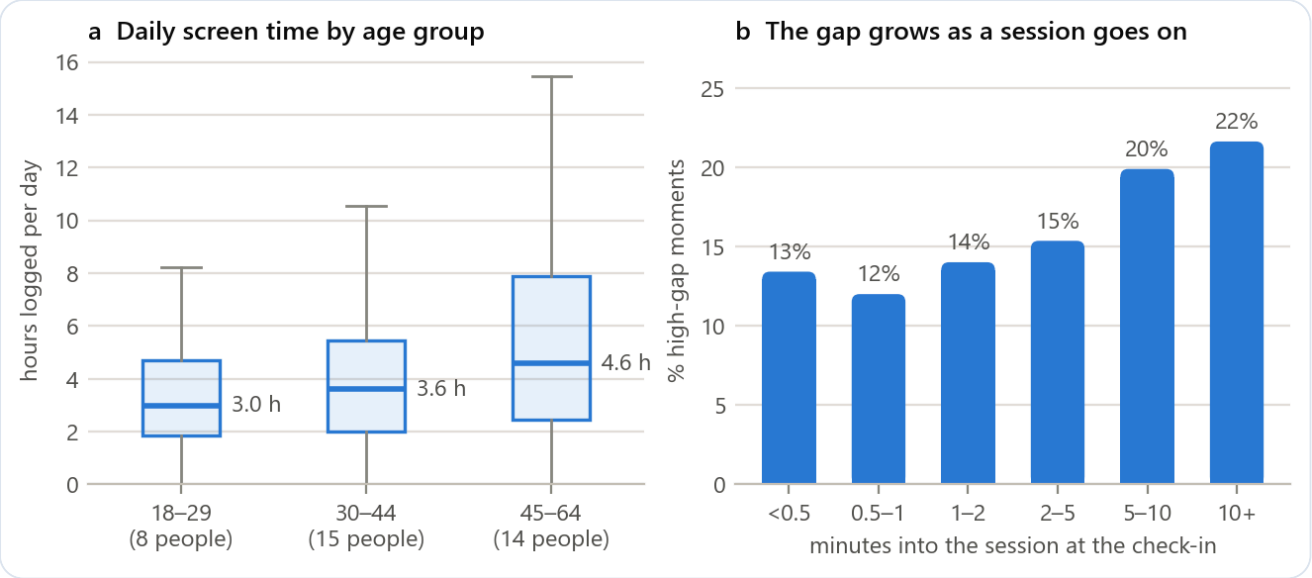


Figure 4. (a) Screen time logged per participant-day, by age group (box: interquartile range; line: median). (b) Share of high-gap moments by minutes already spent in the session at the check-in.

0.44

intraclass correlation: about 44% of the gap's variation lies between people

REPORT: INSIGHT 5 · FIGURE 5

The gap is strongly person-specific. Individual high-gap shares range from 0% to 97.2% (median 7.0%). This explains why one shared model barely works for new users (R^2 = .035; Zheng & Choe, 2026b), and it makes personalization the core of NudgeWell.

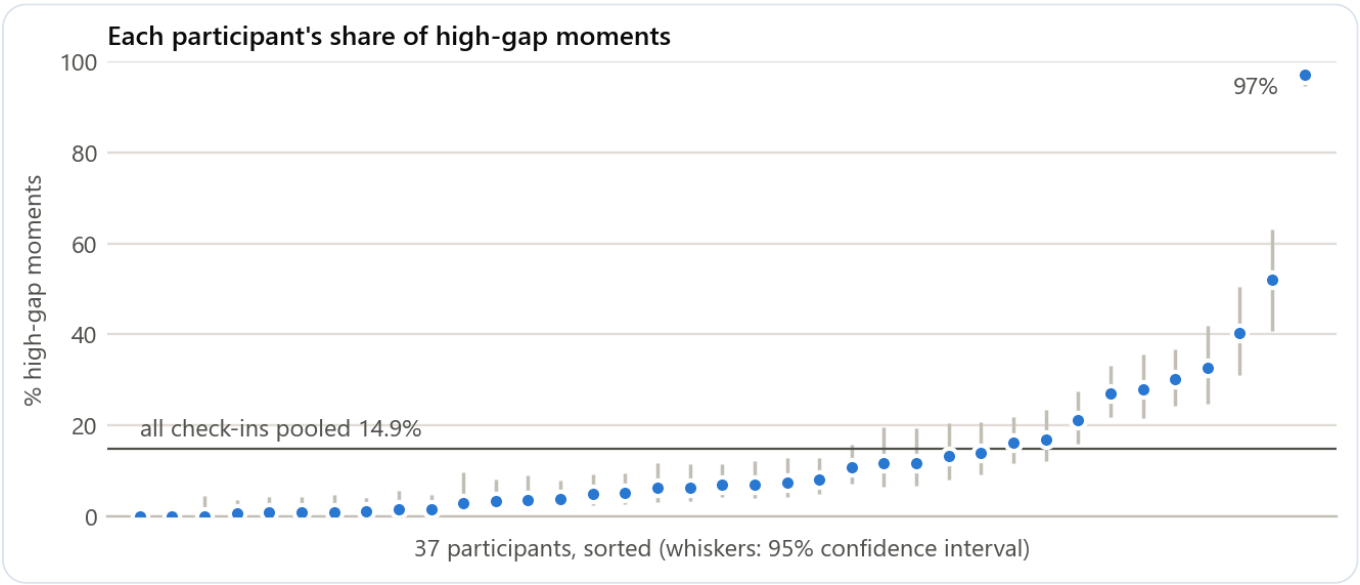


Figure 5. Each participant's share of high-gap moments, sorted, with 95% confidence intervals. Participant codes are omitted.

WHAT THE LITERATURE SAYS

14 studies, four themes, four gaps

Part 1 of the report reviews 14 studies in four themes. Each card gives a theme's main takeaway; the coral cards are the gaps NudgeWell addresses.

MEASUREMENT

Scales measure people, not moments

Self-reported use only moderately matches logged use (Parry et al., 2021). The SAS-SV scale has a validated Turkish version (Noyan et al., 2015).

CONSEQUENCES

Patterns matter more than minutes

In 10,099 US adults, objectively measured use showed weak or null links with mood (Winbush et al., 2025). Lower-well-being profiles were marked by night-time use (Katevas et al., 2018).

INTERVENTIONS

Brief nudges work, then fade

Short frictions made app openings more intentional for 1,039 *one sec* users, who then took breaks and rebounded (Haliburton et al., 2024). *TypeOut* cut app usage by more than half (Xu et al., 2022).

AI

Models find problem users, not moments

Time2Stop learns when to intervene (Orzikulova et al., 2024). Only AwareMind predicts the gap itself, and its model barely works for new users (Zheng & Choe, 2026b).

Measurement

Only one dataset captures the momentary clash between intention and behaviour.

Prediction

The gap has only been modelled with random forests: no sequence models and no systematic personalization.

Intervention

Just-in-time systems decide *when* to interrupt, not whether the person wants to stop.

Population

There is no momentary data from Turkey, or from outside the US, Europe and East Asia.

TECHNOLOGY CHOICES

A deliberately modest stack

Part 3 of the report compares sensing APIs, models and deployment options. These are the choices it makes.

Android usage access

`UsageStatsManager` (Android Developers, n.d.). Events are kept for only a few days, so the app stores its own minimal history.

Baselines

A random forest, as in the paper, plus gradient-boosted trees (XGBoost).

Temporal model

A compact TCN or GRU over recent sessions.

Personal and explainable

Global, personal and combined variants, with calibration, a daily prompt budget and SHAP explanations (Lundberg & Lee, 2017).

On the phone

LiteRT or ONNX Runtime (Google AI Edge, n.d.; ONNX Runtime, n.d.), inside the existing Kotlin/Compose Eşik app.

RULED OUT FOR NOW

AccessibilityService: richer data, but it needs Play Store disclosure and approval, and it is weaker on privacy.

iOS Screen Time APIs: built around parental controls, with no raw logs for custom models.

Transformers: they typically need far more data than 5,772 labelled moments.

Cloud inference: easier to update, but it sends behavioural data off the phone.

DATA QUALITY AND PRIVACY

Known limits, and how we handle them

From Part 2 of the report (Table 4 and Section 2.4).

Target leakage

`answer`, the `remaining_*` columns, `lock_time` and the undocumented `temp_scaled` (a rescaled copy of the outcome) all reveal the target.

Handling: excluded; features use only what is known at the check-in.

Class imbalance

High-gap moments are 14.9% of check-ins.

Handling: precision–recall AUC and F1, class weights and a tuned threshold; never accuracy alone.

Self-report bias

11 participants answered 10 to more than half of their prompts (up to 89%).

Handling: per-person standardization, personal models, and a sensitivity analysis without near-constant raters.

Missing app information

12.4% of check-ins (717) lack the app category; activity is missing for 2.3%.

Handling: explicit "not recorded" categories with missingness flags; nothing is dropped silently.

Repeated measures

72–294 check-ins per person; 5,549 sessions, 204 of them with more than one check-in.

Handling: leave-one-participant-out splits for new users; chronological splits within each person.

Small, US-only sample

37 adults in five US time zones. Group differences, such as women 19.7% against men 8.1%, rest on 22 and 15 people.

Handling: confidence intervals, per-person results, and explicit limits on transfer to Turkish users.

Privacy by design under KVKK (Law No. 6698, 2016)

Explicit opt-in consent

Before any processing, in plain language (Art. 5). An inferred "problematic use" state may count as health data, a special category (Art. 6).

Minimal data

App categories and session timing only: no message or keystroke content and no screen captures (Art. 4).

On the device

Processing and storage stay on the phone: no account, no advertising use, no sale of data (Art. 12).

User control

View, correct, export or delete the data; pause the app or withdraw at any time (Art. 11).

Adults only

Minors are outside the target group; including them would need parental consent and a separate ethics review.

Real help, not therapy

Crisis-related language leads to Yeşilay YEDAM and the 115 line, never to generated advice.

For AwareMind itself, we report only aggregated results, show no participant codes, keep the raw files local, and use the data for coursework analysis only. OSF lists no licence for the data, so we will obtain written permission from the authors before model training or publication.

WHAT COMES NEXT

From review to concept note

Model

Benchmark compact temporal and personalized models of the gap against the published random forest, and report negative results honestly.

Evaluation

Leakage-safe splits: leave-one-participant-out for new users, and chronological splits within each person to test personalization.

Prototype

An opt-in Android prototype built on the Eşik app: quiet by default, a daily prompt budget, and one suggestion from the user's own list.

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Interactive supplement to the Literature, Data & Technology Review, 12 October 2026. Statistics and figures come from the same analysis notebook as Part 2 of the report; AwareMind data are used for coursework analysis only and shown in aggregate.